

5. Mortality

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1 Environment preparation

Run scripts with the needed packages and the raw data frame.

```
source("0a_packages.R")
source("0b_script_data.R")
```

2 Counts and cumulative proportions

2.1 By lineage, tumoral status, and treatment condition

```
df %>%
  group_by(lineage, tum_state, treatment_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 18 x 6
##   lineage tum_state treatment_condition n_total n_dead prop
##   <fct>   <fct>       <fct>             <int> <dbl> <dbl>
```

##	1	HC_MT	Healthy	Control	48	1	2.08
##	2	HC_MT	Healthy	Shaking treatment	48	6	12.5
##	3	HO_MT	Healthy	Control	62	1	1.61
##	4	HO_MT	Healthy	Shaking treatment	38	1	2.63
##	5	HO_MT	Tumoral	Control	10	0	0
##	6	HO_MT	Tumoral	Shaking treatment	34	1	2.94
##	7	HO_SPC	Healthy	Control	62	2	3.23
##	8	HO_SPC	Healthy	Shaking treatment	29	1	3.45
##	9	HO_SPC	Tumoral	Control	10	0	0
##	10	HO_SPC	Tumoral	Shaking treatment	43	0	0
##	11	HO_SPT	Healthy	Control	12	5	41.7
##	12	HO_SPT	Healthy	Shaking treatment	25	3	12
##	13	HO_SPT	Tumoral	Control	60	2	3.33
##	14	HO_SPT	Tumoral	Shaking treatment	47	0	0
##	15	HO_VLN	Healthy	Control	96	1	1.04
##	16	HO_VLN	Healthy	Shaking treatment	96	0	0
##	17	HV_GAL	Healthy	Control	72	1	1.39
##	18	HV_GAL	Healthy	Shaking treatment	72	6	8.33

2.2 By lineage and tumoral status

```
df %>%
  group_by(lineage, tum_state) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 9 x 5
##   lineage tum_state n_total n_dead prop
##   <fct>   <fct>     <int>  <dbl> <dbl>
## 1 HC_MT   Healthy      96      7  7.29
## 2 HO_MT   Healthy     100      2    2
## 3 HO_MT   Tumoral      44      1  2.27
## 4 HO_SPC  Healthy      91      3  3.3
## 5 HO_SPC  Tumoral      53      0    0
## 6 HO_SPT  Healthy      37      8 21.6
## 7 HO_SPT  Tumoral     107      2  1.87
## 8 HO_VLN  Healthy     192      1  0.52
## 9 HV_GAL  Healthy     144      7  4.86
```

2.3 By lineage and treatment condition

```
df %>%
  group_by(lineage, treatment_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
```

```
prop = round(n_dead / n_total * 100, 2),
.groups = "drop")
```

```
## # A tibble: 12 x 5
##   lineage treatment_condition n_total n_dead prop
##   <fct>    <fct>              <int>  <dbl> <dbl>
## 1 HC_MT    Control                48      1  2.08
## 2 HC_MT    Shaking treatment        48      6 12.5
## 3 HO_MT    Control                72      1  1.39
## 4 HO_MT    Shaking treatment        72      2  2.78
## 5 HO_SPC   Control                72      2  2.78
## 6 HO_SPC   Shaking treatment        72      1  1.39
## 7 HO_SPT   Control                72      7  9.72
## 8 HO_SPT   Shaking treatment        72      3  4.17
## 9 HO_VLN   Control               96      1  1.04
##10 HO_VLN   Shaking treatment        96      0   0
##11 HV_GAL   Control                72      1  1.39
##12 HV_GAL   Shaking treatment        72      6  8.33
```

2.4 By tumoral status and treatment condition

```
df %>%
  group_by(tum_state, treatment_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 4 x 5
##   tum_state treatment_condition n_total n_dead prop
##   <fct>    <fct>              <int>  <dbl> <dbl>
## 1 Healthy    Control                352     11  3.12
## 2 Healthy    Shaking treatment        308     17  5.52
## 3 Tumoral    Control                 80      2   2.5
## 4 Tumoral    Shaking treatment        124      1  0.81
```

2.5 By lineage

```
df %>%
  group_by(lineage) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 6 x 4
```

```
## lineage n_total n_dead prop
## <fct>      <int> <dbl> <dbl>
## 1 HC_MT      96      7  7.29
## 2 HO_MT     144      3  2.08
## 3 HO_SPC     144      3  2.08
## 4 HO_SPT     144     10  6.94
## 5 HO_VLN     192      1  0.52
## 6 HV_GAL     144      7  4.86
```

2.6 By tumoral status

```
df %>%
  group_by(tum_state) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 2 x 4
##   tum_state n_total n_dead prop
##   <fct>      <int> <dbl> <dbl>
## 1 Healthy      660      28  4.24
## 2 Tumoral      204       3  1.47
```

2.7 By treatment condition

```
df %>%
  group_by(treatment_condition) %>%
  summarise(
    n_total = n(),
    n_dead = sum(death_state, na.rm = TRUE),
    prop = round(n_dead / n_total * 100, 2),
    .groups = "drop")
```

```
## # A tibble: 2 x 4
##   treatment_condition n_total n_dead prop
##   <fct>              <int> <dbl> <dbl>
## 1 Control            432     13  3.01
## 2 Shaking treatment  432     18  4.17
```